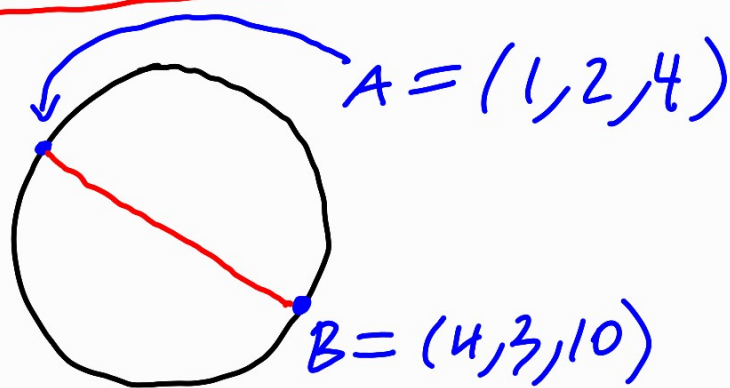


Discussion site contains logistics info, course structure, discussion materials (worksheet, solutions, discussion notes), and other info for this class

Anonymous feedback survey in 3 ± 1 weeks, timing TBD

2: Sphere, ^{diameter} endpoints $(1, 2, 4)$, $(4, 3, 10)$.

Recall that we need the center & radius to find the equation.



Since AB is a diameter, we have
 $r = \frac{1}{2}d = \frac{1}{2}|\vec{AB}|$

$$= \frac{1}{2}|B - A| = \frac{1}{2}|\langle 3, 1, 6 \rangle| =$$

$$\frac{1}{2}\sqrt{3^2 + 1^2 + 6^2} = \frac{1}{2}\sqrt{46} \Rightarrow r^2 = \frac{23}{2}$$

Recall center is midpoint of any diameter.

So $C = \frac{1}{2}(A+B) = \frac{1}{2}(5, 5, 14) = (\frac{5}{2}, \frac{5}{2}, 7)$. Plug C & r^2 into formula:

$$(x - \frac{5}{2})^2 + (y - 2.5)^2 + (z - 7)^2 = \frac{23}{2}.$$

4: Length 5, opposite $\langle 9, -6, -2 \rangle$

Vectors of opposite/same direction are parallel. Parallel (non-zero) vectors are constant multiples of each other. So we seek a vector $c \langle 9, -6, -2 \rangle$.

Length is 5, so

$$5 = |c \langle 9, -6, -2 \rangle| = |c| \sqrt{81 + 40} = 11|c|$$

$$\Rightarrow |c| = \frac{5}{11} \Rightarrow c = \pm \frac{5}{11}$$

$c > 0 \Rightarrow$ same dir, $c < 0 \Rightarrow$ oppos. dir,
so in this case $c < 0 \Rightarrow c = -\frac{5}{11}$

$\Rightarrow$ the vector is $-\frac{5}{11} \langle 9, -6, -2 \rangle$,
or $\langle -\frac{45}{11}, \frac{30}{11}, \frac{10}{11} \rangle$.

7: θ between $a = \langle 2, 0, -3 \rangle$, $b = \langle -2, 3, 1 \rangle$

Recall: $\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}$.

$$a \cdot b = 2(-2) + 0 \cdot 3 + (-3) \cdot 1 = -4 - 3 = -7$$

$$\|a\| = \sqrt{2^2 + 0^2 + 3^2} = \sqrt{13}, \quad \|b\| = \sqrt{14}$$

$$\text{So } \cos \theta = \frac{-7}{\sqrt{13}\sqrt{14}} = -\frac{\sqrt{7}}{\sqrt{26}}, \quad \text{so}$$

$$\theta = \cos^{-1}(\cos \theta) = \cos^{-1}\left(-\frac{\sqrt{7}}{\sqrt{26}}\right)$$

$$= \cos^{-1}\left(-\frac{\sqrt{182}}{26}\right)$$

8: $a = \langle -1, 4, 8 \rangle$, $b = \langle 18, 2, 1 \rangle$, find scalar & vector proj. of b onto a .

(2) Recall scalar projection formula:

$$\text{scproj} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{-18 + 8 + 8}{\sqrt{1 + 16 + 64}} = \frac{-2}{\sqrt{81}} = -\frac{2}{9}$$

(b) Recall vector projection onto $\vec{a}$ is just $(\text{scproj}) \cdot (\text{unit vector of } \vec{a})$

$$\text{unit vector is } \frac{\vec{a}}{|\vec{a}|} = \frac{\langle -1, 4, 8 \rangle}{9} = \frac{1}{9} \langle -1, 4, 8 \rangle$$

$$\text{so } \text{proj}_{\vec{a}} \vec{b} = -\frac{2}{9} \cdot \frac{1}{9} \langle -1, 4, 8 \rangle = \boxed{-\frac{2}{81} \langle -1, 4, 8 \rangle}$$

9: $F = \langle 8, -6, 5 \rangle$ on object from $(0, 6, 4)$ to $(4, 14, 22)$ along straight line, find work done.

Recall: $W = \vec{F} \cdot \vec{d}$ where $\vec{F}$ is force vector & $\vec{d}$ displacement of the object, for line + constant force.

$$\begin{aligned} d &= \text{change in position} = \text{end} - \text{start} \\ &= \langle 4, 14, 22 \rangle - \langle 0, 6, 4 \rangle = \langle 4, 8, 18 \rangle \end{aligned}$$

So $F \cdot d = \langle 8, -6, 5 \rangle \cdot \langle 4, 8, 18 \rangle$
 $= 8 \cdot 4 - 6 \cdot 8 + 5 \cdot 18 = 32 - 48 + 90 = 74,$
 so $W = 74 \text{ N} \cdot \text{m} = 74 \text{ J}.$

6: a, b, c are vectors, which of $(a \cdot b) \cdot c$,
 $(a + b) \cdot c$, $a + (b \cdot c)$ meaningful

(a) $(a \cdot b) \cdot c = \text{scalar} \cdot \text{vector}$
 $= \begin{cases} \text{meaningful if 1st "." is mult.} \\ \text{invalid if 1st "." is dot product} \end{cases}$

(b) $(a + b) \cdot c = \text{vector} \cdot \text{vector} = \text{scalar},$
 so this expression makes sense

(c) $a + (b \cdot c) = \text{vector} + \text{scalar},$ so
 this does not make sense.

5: orthogonal, parallel, or neither

(a) $a = \langle 4, -1, 4 \rangle$, $b = \langle 5, 12, -2 \rangle$. Let's
 check dot product.

$a \cdot b = 20 - 12 - 8 = 0$, so orthogonal.

(b) $a = \langle 6, 5, -2 \rangle$, $b = \langle 5, 0, 9 \rangle$.

$a \cdot b = 30 + 0 - 18 = 12 \neq 0$, so not perp.

If a & b are parallel, then there exists c such that $\vec{a} = c \cdot \vec{b} = \langle 5c, 0, 9c \rangle$

Take the 2nd component of both sides

$\Rightarrow 5 = 0$, impossible.

Not perp. & not parallel $\Rightarrow$ neither

(c) $a = \langle -18, 15 \rangle$, $b = \langle 12, -10 \rangle$.

Let's try parallelity again:

$\langle -18, 15 \rangle = \vec{a} = c \cdot \vec{b} = \langle 12c, -10c \rangle$.

1st component gives $-18 = 12c \Rightarrow c = -\frac{3}{2}$.

This also works on 2nd one, so $\vec{a} = -1.5 \vec{b}$,

so a & b are parallel.

1: $P = (3, 4, 5)$, proj onto xy & xz -planes

(a) Recall xy -plane is the plane $z=0$,

$P = (3, 4, 5)$ so the projection is $(3, 4, 0)$.



(b) xz-plane is plane $y=0$, so it's $(3, 0, 5)$.

1c: $|O-P| = \sqrt{\dots} = 5\sqrt{2}$

d: $P-O = \langle 3, 4, 5 \rangle = 3i + 4j + 5k$

3a: $\langle 16, -8, -2 \rangle$ b: $\langle 20, -24, 30 \rangle$

c: $\sqrt{130}$ d: 18

4: $|c \langle 9, -6, -2 \rangle| = 5 \Rightarrow c = \pm \frac{5}{11}$,

opp. $\Rightarrow c = -\frac{5}{11} \Rightarrow -\frac{5}{11} \langle 9, -6, -2 \rangle$

8: a: $\frac{z \cdot b}{|z|} = \frac{-2}{9}$

b: $\frac{z \cdot b}{|z|} \cdot \frac{z}{|z|} = -\frac{2}{9} \cdot \frac{1}{9} z = -\frac{2}{81} \langle -1, 4, 8 \rangle$

9: $d = (4, 14, 22) - (0, 6, 4) = (4, 8, 18)$,

$W = F \cdot d = 74 \text{ N} \cdot \text{m} = 74 \text{ J}$
